CHAPTER 1

INTRODUCTION

1.1 Overview

Traffic congestion at road intersections is a common problem in urban and semi-urban areas. Conventional traffic signals generally operate using fixed timing plans, where each road receives a predetermined green interval regardless of the actual traffic present. When traffic density is unequal, this approach causes unnecessary waiting on lightly occupied roads and insufficient green time for heavily occupied roads.

The proposed Density Based Traffic Light Control System is a four-approach embedded prototype that estimates traffic density on the North, South, East and West roads using three IR sensing zones per approach. An Arduino Mega 2560 reads all twelve sensors, classifies the density of each approach, selects the most demanding road, and adjusts the green signal duration according to the detected density. A 16 x 2 I2C LCD displays live density and signal information, while red, yellow and green LEDs represent the traffic lights.

The project demonstrates adaptive signal control using low-cost hardware. It is intended as an educational prototype of a smart intersection rather than a road-certified traffic controller. The system focuses on density estimation, safe signal sequencing, fair priority handling and visual monitoring.

1.2 Problem Statement

A fixed-time traffic controller cannot respond to changing traffic demand. One road may be empty while another has a long queue, yet both can receive the same green time. This leads to inefficient use of intersection capacity, additional delay and poor demonstration of real-time control.

Problem	Conventional Approach	Proposed Solution
Unequal traffic density	Same or fixed green duration	Green duration changes with detected density
Empty road receives time	Signal sequence may continue even with no vehicles	Empty approaches are skipped
High-density road waits longer	No dynamic priority	Highest detected density is served first
Equal density tie	Fixed priority can favor one road	Rotating tie priority improves fairness
Limited status visibility	No local traffic status display	LCD shows density, phase and timer
Unsafe direct switching	Improper prototype logic may switch directly	Yellow phase and all-red safety gap are enforced

1.3 Objectives of the Project

Primary Objectives

#	Objective
1	Design a four-approach traffic signal prototype using Arduino Mega 2560.
2	Use twelve IR sensors to estimate traffic density on North, South, East and West approaches.
3	Classify traffic as EMPTY, LOW, MEDIUM or HIGH using the number of active sensing zones.
4	Provide adaptive green timing of 7 s, 10 s and 15 s for LOW, MEDIUM and HIGH density.
5	Provide a 3 s yellow interval and 1 s all-red safety interval between green phases.

#	Objective
6	Prioritize higher-density approaches while ensuring fair handling of equal-density roads.
7	Display live density and signal information on a 16 x 2 I2C LCD.

Secondary Objectives

- Filter IR sensor readings using repeated sampling and majority decision to reduce unstable detection.
- Ensure that only one approach can display GREEN at any time.
- Provide Serial Monitor output for testing and debugging.
- Develop a low-cost prototype suitable for academic demonstration and embedded-systems learning.
- Use a service-cycle method so every occupied road receives one turn before a fresh scheduling cycle begins.

1.4 Literature Survey

1.4.1 Traffic Signal Control Methods

Traffic signal control can be broadly divided into fixed-time, vehicle-actuated and adaptive methods. Fixed-time controllers use predetermined phase durations. Vehicle-actuated systems use detectors to extend or request phases based on traffic presence. More advanced adaptive systems continuously estimate demand and adjust timing or phase order. The present project follows the adaptive concept at prototype scale by using IR sensing zones and density-dependent green timing.

Technique	Working Principle	Typical Limitation in Student Prototypes
Fixed-time control	Predetermined phase sequence and timing	Cannot respond to changing traffic demand
Vehicle-actuated control	Presence detector requests or extends a phase	Often handles presence but not multiple density levels
Camera/image processing	Density estimated from images	Higher processing, lighting and software complexity
IR sensor based control	Discrete sensing zones estimate occupancy	Range and alignment depend on sensor calibration
Proposed method	Three IR zones per road + adaptive timing + fair priority	Prototype-scale density estimation, not exact vehicle counting

1.4.2 IR Sensing for Density Estimation

IR obstacle sensors provide a simple digital indication when an object is within the calibrated detection range. In this project, three sensors are assigned to each road. The final program does not require a strict sensor order: any one active sensor represents LOW density, any two active sensors represent MEDIUM density and all three active sensors represent HIGH density. This makes the physical model more tolerant to toy-car placement and sensor alignment.

Active Sensors	Possible Patterns	Density
0	000	EMPTY
1	100, 010, 001	LOW
2	110, 101, 011	MEDIUM
3	111	HIGH

1.4.3 Gap and Project Contribution

- A simple fixed sequence does not demonstrate density-aware control.
- A single sensor per road indicates presence but cannot represent multiple density levels.
- Fixed tie priority can repeatedly favor one approach when traffic levels are equal.
- The developed prototype combines multi-zone sensing, adaptive timing, one-green-at-a-time safety, fair tie handling and LCD monitoring in one integrated system.

1.5 Existing and Proposed System

1.5.1 Existing System

Component / Function	Existing Approach
Signal timing	Fixed or predetermined duration
Traffic input	No density input in basic fixed-time model
Priority	Fixed sequence
Empty road handling	May still receive a green phase
Status display	Usually not available in a simple prototype

1.5.2 Proposed System

Component / Function	Proposed Approach
Controller	Arduino Mega 2560
Traffic sensing	12 IR sensors, 3 per approach
Density levels	EMPTY / LOW / MEDIUM / HIGH
Adaptive green time	LOW 7 s, MEDIUM 10 s, HIGH 15 s
Priority	Highest density first; rotating selection for ties
Safety sequence	GREEN -> YELLOW 3 s -> ALL RED 1 s -> next road
Display	16 x 2 I2C LCD with density, signal and raw sensor information

Advantages of the Proposed System:

- Responds to actual detected density instead of relying only on fixed timing.
- Skips empty approaches in the active service cycle.
- Provides fairer handling when two or more roads have equal density.
- Uses inexpensive and easily available components.
- Provides clear visual feedback through traffic LEDs, LCD and Serial Monitor.

1.6 Hardware and Software Components Used

1.6.1 Hardware Components Summary

S.No	Component	Quantity	Purpose
1	Arduino Mega 2560	1	Main controller
2	IR obstacle sensor modules	12	Three traffic sensing zones per approach
3	Red LEDs	4	STOP indication
4	Yellow LEDs	4	Transition / caution indication

S.No	Component	Quantity	Purpose
5	Green LEDs	4	GO indication
6	220 ohm resistors	12	LED current limiting
7	16 x 2 I2C LCD	1	Live system display
8	Breadboards	As required	Prototype wiring
9	Jumper wires	Set	Electrical connections
10	5 V USB / regulated supply	1	Power for controller and low-power modules
11	Road-model base / foam board / plywood	1	Physical intersection model
12	Toy vehicles	As required	Traffic demonstration

1.6.2 Software Components

Software / Library	Purpose
Arduino IDE	Program development, compilation, upload and Serial Monitor
Wire.h	I2C communication
LiquidCrystal_I2C.h	16 x 2 LCD control
string.h	String length support for clean LCD line writing

1.7 Organization of the Project

Chapter	Title	Description
1	Introduction	Overview, problem statement, objectives, survey, components
2	System Design and Operation	Block diagram, density logic, priority and timing
3	Hardware Design and Description	Controller, sensors, LEDs, LCD and power
4	Hardware Implementation	Complete wiring, pin tables and assembly
5	Software Description	Libraries, data structures, functions and state machine
6	Software Implementation	Flowcharts, algorithms and program logic
7	Experimental Results	Functional tests, timing tests and observed outputs
8	Applications, Advantages and Disadvantages	Use cases, benefits and limitations
9	Conclusion and Future Scope	Summary, achievements and improvements

1.8 Summary

This chapter introduced the need for density-based traffic control, defined the project objectives, compared the proposed method with a fixed-time system, and listed the hardware and software used. The following chapter explains the system architecture and complete operating mechanism.

CHAPTER 2

SYSTEM DESIGN AND OPERATION

2.1 Block Diagram

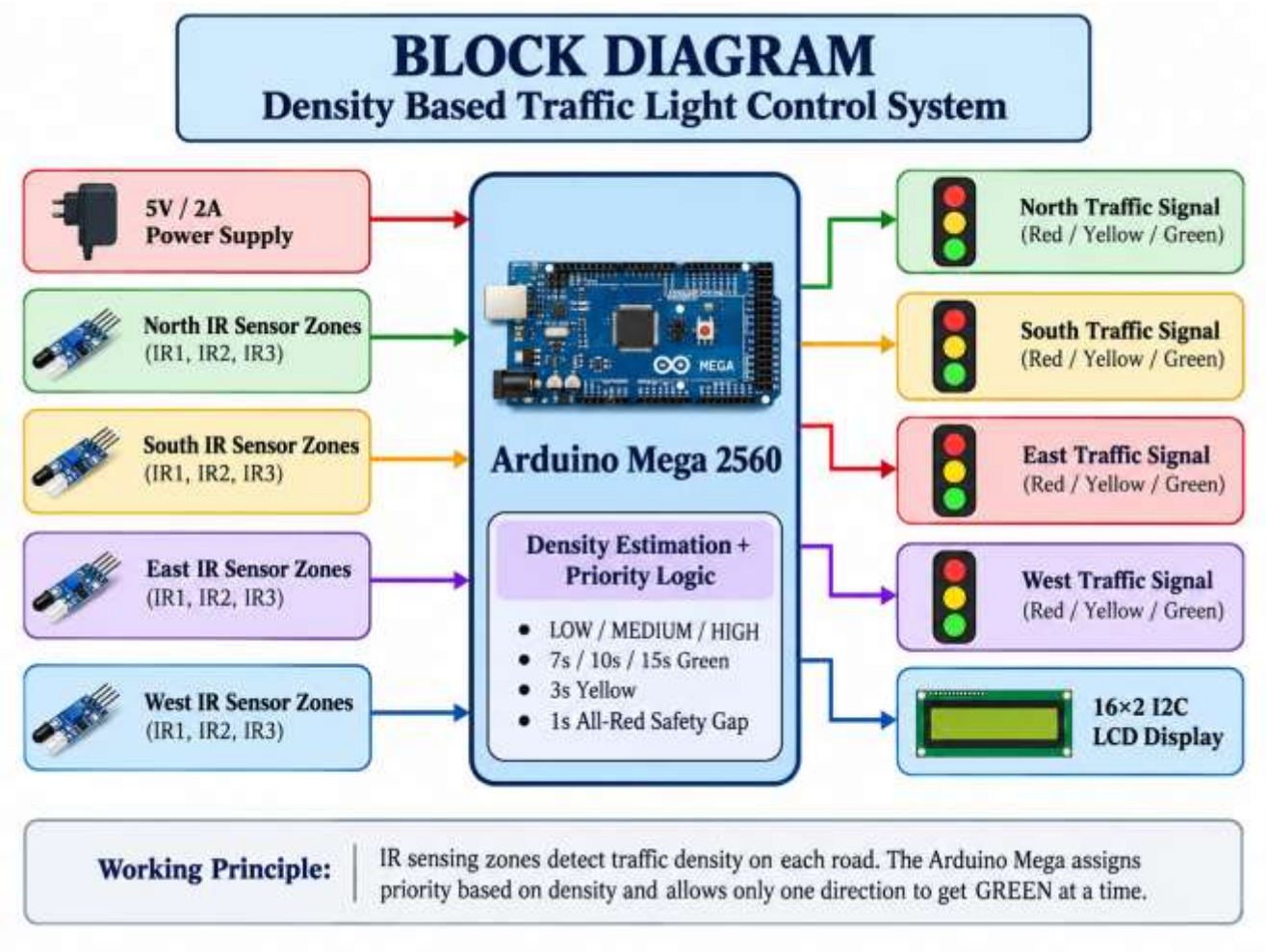


Figure 2.1: System block diagram of the density based traffic light controller

2.2 Working Mechanism

2.2.1 Sensor Acquisition

Each of the four approaches contains three IR sensors. The Arduino Mega reads all twelve digital outputs. Most of the obstacle sensor modules used in the prototype are active-LOW; therefore a LOW digital input is interpreted as object detected. To reduce unstable switching, each sensor is sampled seven times and is considered active when at least four samples indicate detection.

2.2.2 Density Classification

After filtering, the program counts the number of active sensors on each approach. The order of sensor activation does not affect the result. This method was selected for the final physical model because toy vehicles may not occupy perfectly continuous positions.

Number of Active Sensors	Density	Green Duration
0	EMPTY	No green phase requested
1	LOW	7 seconds

Number of Active Sensors	Density	Green Duration
2	MEDIUM	10 seconds
3	HIGH	15 seconds

2.2.3 Service Cycle and Priority

At the beginning of a service cycle, the current density of all four approaches is captured. Empty approaches are excluded. The controller then chooses the highest-density unserved approach. After that approach completes GREEN, YELLOW and ALL-RED phases, it is marked as served. The controller continues with the highest-density unserved approach until every occupied road in that captured cycle has received one turn.

When two or more approaches have the same highest density, a rotating priority is used. The search begins after the most recently served approach. This prevents a fixed lane from always winning equal-density ties and improves fairness in the four-road prototype.

2.2.4 Signal Phase Sequence

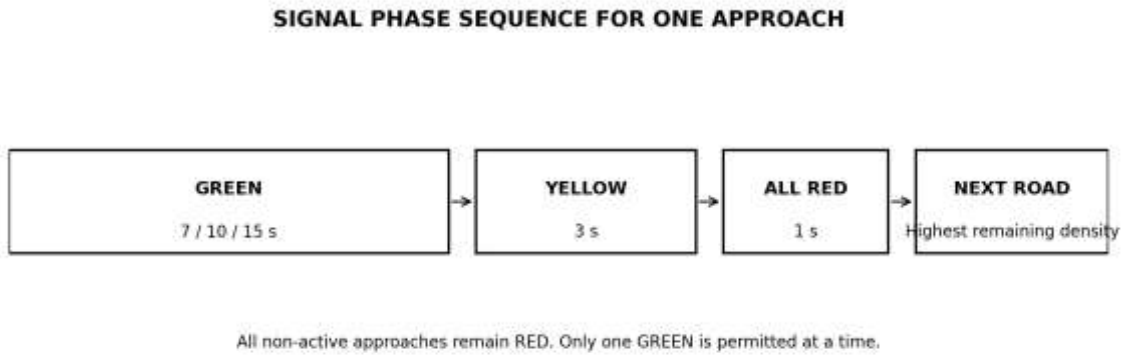


Figure 2.2: Signal phase sequence for each served approach

During a green phase, all other approaches remain RED. After the selected green time expires, the same approach changes to YELLOW for 3 seconds. The controller then forces ALL RED for 1 second before the next road can receive GREEN. This sequence prevents two approaches from being green simultaneously.

2.2.5 LCD Monitoring

The 16 x 2 LCD rotates through multiple information pages. One page shows the overall density status and active signal timer. The other pages show the raw three-sensor patterns and density names for North/South and East/West. This allows the user to verify both the detected input pattern and the decision made by the controller.

LCD Page	Example	Meaning
Overall	N:H S:L E:M W:0 / N GREEN 15s	All road densities + active signal
North/South	N:101 MED / S:001 LOW	Raw sensor patterns and classification
East/West	E:110 MED / W:111 HIGH	Raw sensor patterns and classification

2.3 Operating Scenarios

Scenario	System Response
All approaches EMPTY	All signals remain RED and the system waits for traffic.
One occupied approach	Only that approach receives green according to its density.
Mixed densities	HIGH roads are served before MEDIUM, then LOW.
Equal densities	Rotating tie selection provides fair service.
All roads HIGH	All four roads receive a 15 s green turn with yellow and all-red transitions.

2.4 Sensor Placement

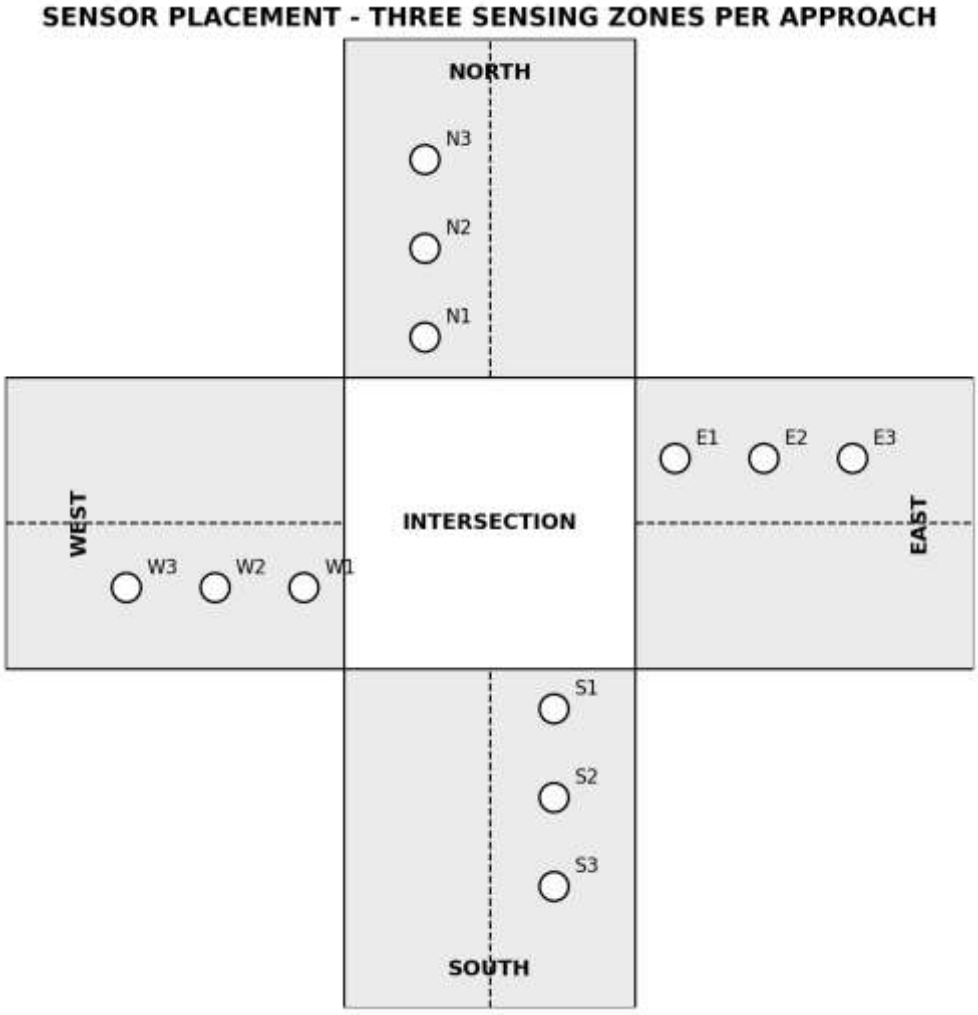


Figure 2.3: Three sensing zones on each of the four approaches

The physical labels IR1, IR2 and IR3 are still useful for wiring and troubleshooting, even though the final density algorithm is order-independent. The sensors should be positioned so that toy vehicles can be placed within clearly separated detection zones without neighboring sensors interfering with each other.

2.5 Summary

The system combines filtered IR sensing, four-level density classification, density-aware scheduling, fair tie handling, adaptive green timing, safe yellow/all-red transitions and live LCD monitoring. The next chapter describes the hardware modules used to implement this design.

CHAPTER 3

HARDWARE DESIGN AND DESCRIPTION

3.1 Arduino Mega 2560

The Arduino Mega 2560 is used as the central controller because the final prototype requires twelve IR inputs, twelve LED outputs and two I2C lines for the LCD. A smaller controller would require additional I/O expansion hardware. The Mega provides sufficient digital pins for direct connection and simplifies troubleshooting.

Specification	Value / Use in Project
Microcontroller	ATmega2560
Operating logic	5 V
Clock frequency	16 MHz
Digital I/O	54 digital pins (sufficient for 24 sensor/LED signals plus LCD)
Analog inputs	16
Flash memory	256 KB
SRAM	8 KB
Project role	Sensor acquisition, density calculation, priority scheduling, signal control and LCD update

3.2 IR Obstacle Sensor Modules

Twelve digital IR obstacle sensor modules are used, three for each approach. Each module has VCC, GND and digital OUT connections and includes an adjustable potentiometer for setting the detection threshold. The modules used in the prototype produce an active-LOW output when an object is detected.

Parameter	Project Use
Supply	5 V rail used in the prototype
Output	Digital detection signal to Arduino Mega
Quantity	12
Calibration	Potentiometer adjusted for stable toy-vehicle detection
Filtering	7 digital samples; active if at least 4 indicate detection

3.3 Traffic Signal LEDs

Each approach has a red, yellow and green LED, giving a total of twelve LEDs. A current-limiting resistor is connected in series with every LED. The software ensures that the inactive approaches remain red while only the selected approach can change to green or yellow.

LED	Quantity	Function
Red	4	STOP / inactive approach
Yellow	4	Transition before red
Green	4	Selected approach allowed to proceed

3.4 16 x 2 I2C LCD

A 16 x 2 character LCD with an I2C backpack is used for local monitoring. The tested address is 0x27. The I2C interface reduces the display wiring to VCC, GND, SDA and SCL. On the Arduino Mega, SDA is pin 20 and SCL is pin 21.

Parameter	Value / Connection
Display	16 characters x 2 rows
Interface	I2C
Tested address	0x27
SDA	Mega pin 20
SCL	Mega pin 21
Displayed information	Density, active signal, timer and raw sensor patterns

3.5 Power and Prototyping Hardware

The system is powered from a stable 5 V source / USB supply during testing. Breadboards and jumper wires are used for the prototype. All modules must share a common ground. For a final physical model, wiring should be routed and labelled to reduce accidental disconnection.

Item	Purpose
5 V regulated / USB supply	Controller and low-power electronics
Breadboards	Temporary interconnection of LEDs and power rails
Jumper wires	Sensor, LED, LCD and power connections
Foam board / plywood base	Physical four-way road model
Toy vehicles	Demonstration of different traffic densities

3.6 Summary

The hardware was selected to provide enough direct I/O for the complete four-road prototype while keeping the design inexpensive and easy to demonstrate. The next chapter gives the exact wiring and implementation details.

CHAPTER 4

HARDWARE IMPLEMENTATION

4.1 Complete System Wiring

The final hardware uses one Arduino Mega 2560, twelve IR sensor inputs, twelve traffic LED outputs and an I2C LCD. All sensor grounds, LED grounds and LCD ground are connected to the common system ground.

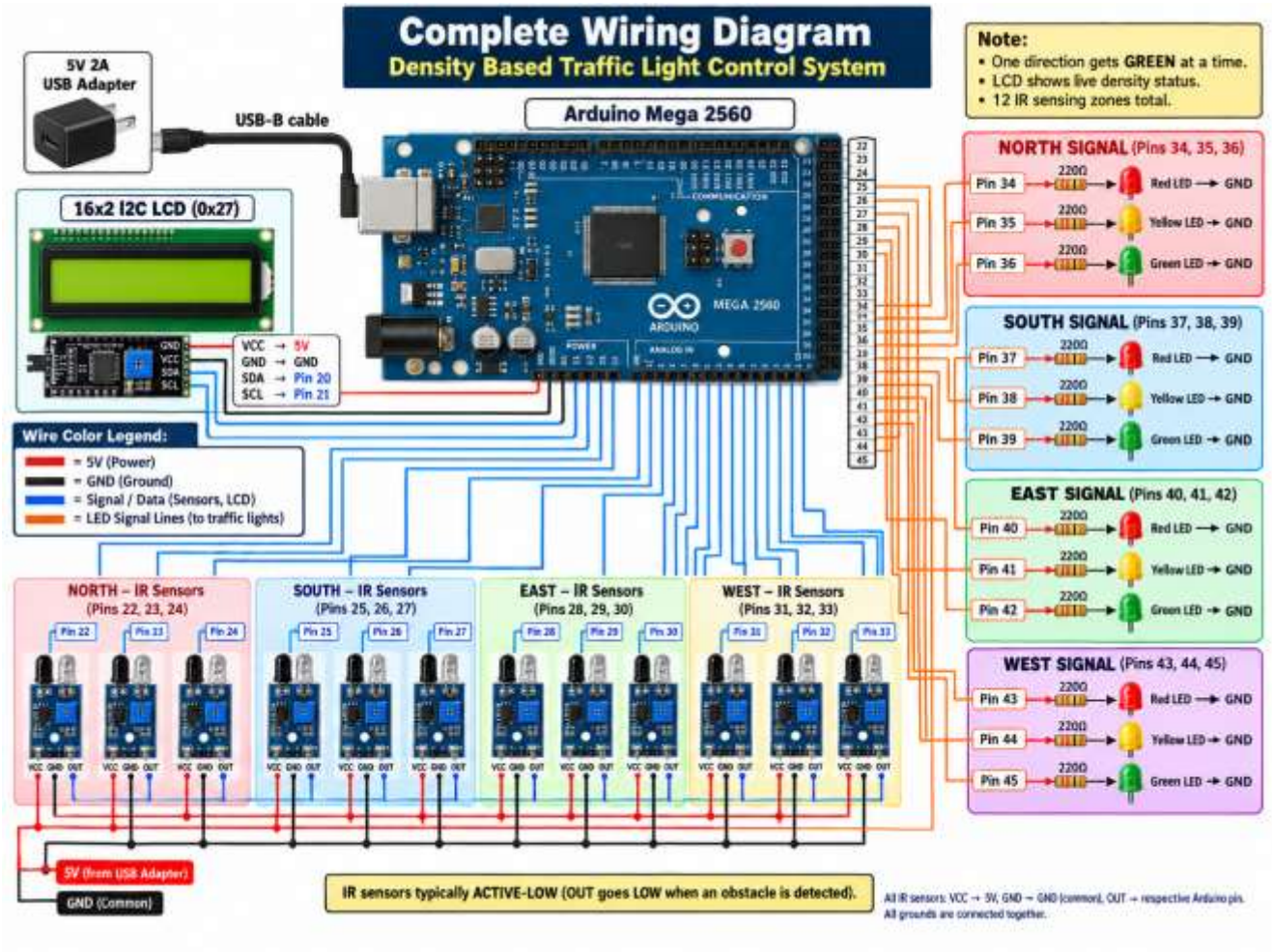


Figure 4.1: Functional wiring overview

4.2 IR Sensor Wiring Table

Approach	IR1	IR2	IR3
North	22	23	24
South	25	26	27
East	28	29	30
West	31	32	33

For every IR sensor: VCC -> 5 V, GND -> GND, OUT -> corresponding Arduino Mega pin. The final density algorithm counts active sensors; therefore IR1/IR2/IR3 order is not required for LOW/MEDIUM/HIGH classification, although labels should be retained for maintenance.

4.3 Traffic LED Wiring Table

Approach	RED	YELLOW	GREEN
North	34	35	36
South	37	38	39
East	40	41	42
West	43	44	45

Each LED is connected through a current-limiting resistor. The LED long leg (anode) is connected toward the Arduino output through the resistor and the short leg (cathode) is connected to GND.

4.4 LCD Wiring

LCD Pin	Arduino Mega Connection
VCC	5 V
GND	GND
SDA	Pin 20
SCL	Pin 21

4.5 Sensor Placement and Calibration

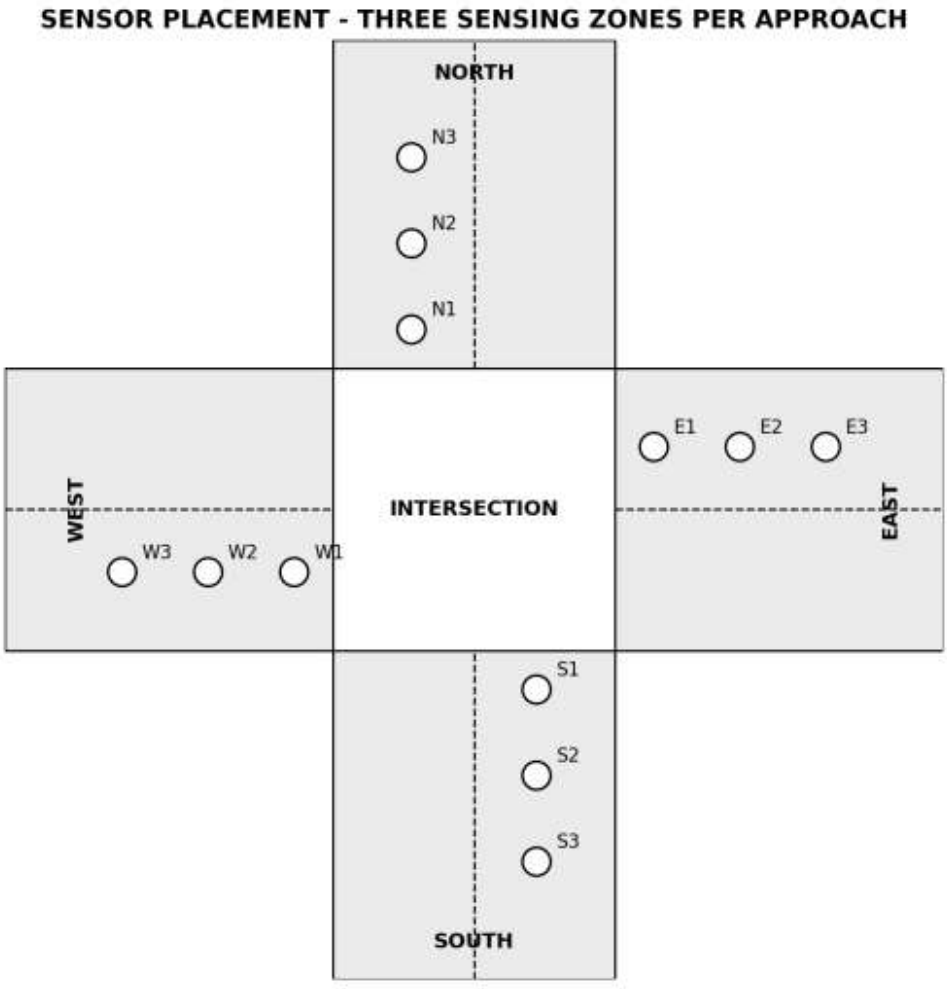


Figure 4.2: Recommended sensor-zone placement on the four-way model

1. Mount three sensors along each approach so that toy vehicles can be placed in distinct sensing zones.
2. Power the system and observe each sensor module indicator LED.
3. Adjust the onboard potentiometer until a stationary toy vehicle produces a stable detection.
4. Verify that each sensor changes independently without continuously triggering an adjacent zone.
5. Label all twelve signal wires and sensor outputs before permanent mounting.

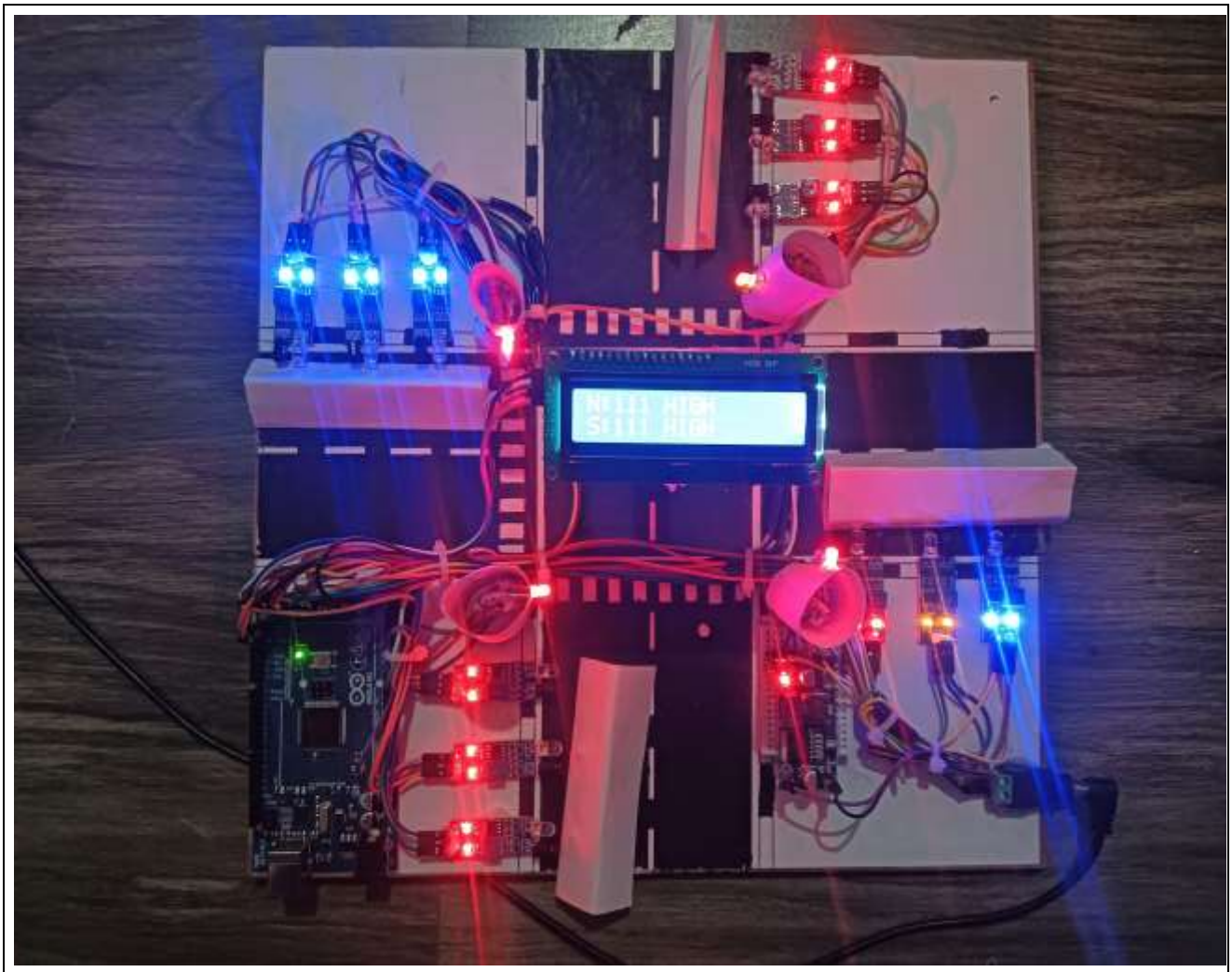
4.6 Assembly Procedure

1. Test the Arduino Mega independently using the onboard LED.
2. Test one traffic signal (North) using pins 34, 35 and 36.
3. Test one IR sensor on pin 22 and calibrate it.
4. Integrate the three North sensors and verify density classification.
5. Test the I2C LCD independently at address 0x27.
6. Integrate North and South, then add East, and finally add West.
7. Verify all twelve IR sensors and all twelve traffic LEDs before mounting on the road model.
8. Perform final mixed-density and all-high traffic demonstrations.

4.7 Power and Wiring Precautions

- Use a stable 5 V source and a common ground for all sensors, LCD and Arduino.
- Do not connect LEDs directly without current-limiting resistors.
- Avoid loose breadboard connections and long unlabelled jumper bundles.
- Calibrate IR sensors after mounting because detection distance can change with angle and target surface.
- If a separate regulated sensor supply is used, connect its ground to Arduino GND.

4.8 Final Hardware Setup Photo



4.9 Summary

The complete hardware implementation uses direct Mega I/O mapping, which simplifies debugging and makes every approach electrically identical. Proper sensor calibration and clear wiring labels are important for reliable demonstration.

CHAPTER 5

SOFTWARE DESCRIPTION

5.1 Arduino IDE

Arduino IDE is used to write, compile and upload the program to the Arduino Mega 2560. The Serial Monitor is used during testing to observe the raw three-sensor pattern and the calculated traffic density for each approach.

5.2 Libraries Used

Library	Purpose
Wire.h	I2C communication with the LCD
LiquidCrystal_I2C.h	LCD initialization, cursor positioning and text output
string.h	String length used when writing fixed-width LCD lines

5.3 Important Data Structures

Variable / Structure	Purpose
sensorPins[4][3]	Stores twelve IR input pin numbers
ledPins[4][3]	Stores red/yellow/green output pins for four approaches
sensorState[4][3]	Stores filtered active/inactive state of each sensor
density[4]	Current density level of each approach
cycleDensity[4]	Density snapshot used for one service cycle
served[4]	Tracks which approaches have already received a turn
activeLane	Stores the approach currently receiving green/yellow
lastServed	Used to rotate equal-density tie priority

5.4 Key Functions

Function	Description
readStableSensor(pin)	Takes seven digital samples and returns detection if at least four are active
readAllSensors()	Updates the filtered states of all twelve IR sensors
calculateLaneDensity(lane)	Counts active sensors and returns 0, 1, 2 or 3
calculateAllDensities()	Calculates density for North, South, East and West
startNewCycle()	Copies current densities and resets served flags
chooseNextLane()	Selects highest-density unserved road and handles ties fairly
getGreenTime(d)	Maps LOW/MEDIUM/HIGH to 7/10/15 seconds
trafficController()	Runs WAITING, GREEN, YELLOW and ALL-RED state transitions
allRed(), setGreen(), setYellow()	Controls safe LED outputs
updateLCD()	Displays overall density, timer and detailed sensor pages
printSerial()	Outputs sensor patterns and density to Serial Monitor

5.5 Traffic State Machine

State	Operation	Exit Condition
WAITING	All signals RED, snapshot current densities and select next road	Valid occupied road selected
GREEN_STATE	Selected road GREEN; others RED	Adaptive green duration expires
YELLOW_STATE	Selected road YELLOW; others RED	3 seconds expire
ALL_RED_STATE	All four approaches RED	1 second expires; select next unserved road

5.6 LCD Display Pages

Page	Line 1	Line 2
0	N:H S:L E:M W:0	N GREEN 15s / ALL RED 1s / waiting
1	N:101 MED	S:001 LOW
2	E:110 MED	W:111 HIGH

5.7 Summary

The program is structured around filtered sensor acquisition, density calculation, a service-cycle scheduler and a millis-based state machine. This separates sensing from signal timing and allows the LCD and sensor processing to continue while a signal phase is active.

CHAPTER 6

SOFTWARE IMPLEMENTATION

6.1 Overall Flowchart

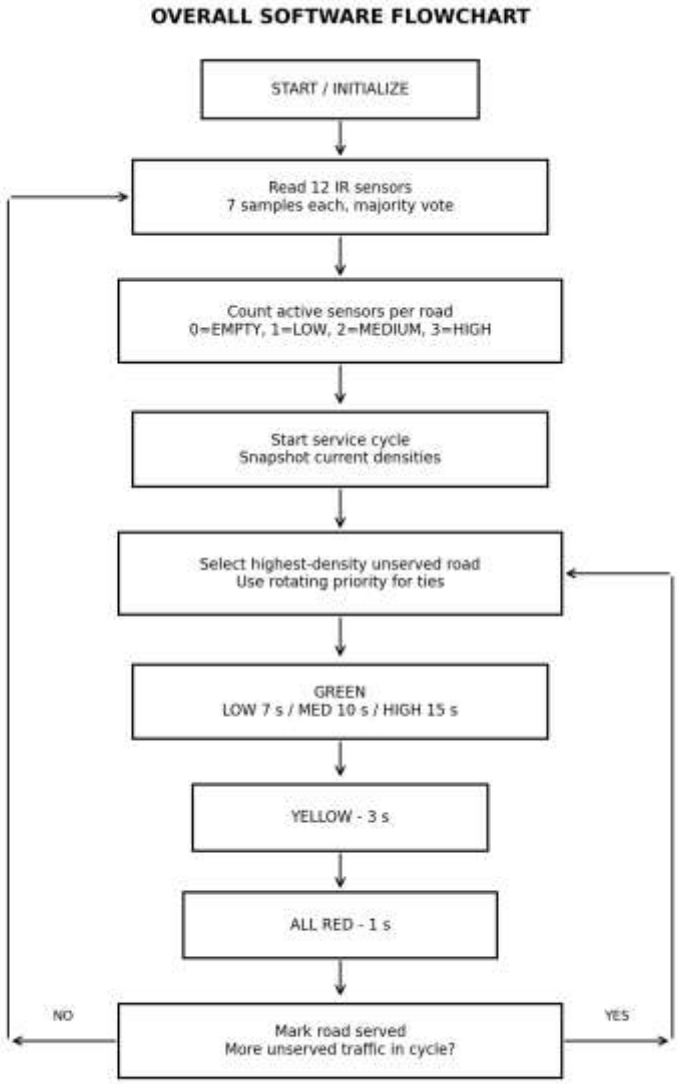


Figure 6.1: Overall program flowchart

6.2 Sensor Filtering Algorithm

1. Select one IR input pin.
2. Read the pin seven times.
3. Interpret LOW as detected for the sensor modules used in the prototype.
4. Count how many of the seven samples indicate detection.
5. If at least four samples indicate detection, store the sensor as ACTIVE; otherwise store it as CLEAR.
6. Repeat for all twelve sensors.

6.3 Density Classification Algorithm

For each road, the program counts the three Boolean sensor states. The returned count is directly used as the density level. This produces an order-independent mapping and eliminates the previous error condition for patterns such as 101 or 011.

Raw Pattern	Active Count	Density
000	0	EMPTY
100 / 010 / 001	1	LOW
110 / 101 / 011	2	MEDIUM
111	3	HIGH

6.4 Priority Selection Algorithm

1. At the start of a service cycle, copy the current density of each occupied road to cycleDensity[].
2. Set served[] = false for all four approaches.
3. Find the highest density among roads that have not yet been served.
4. Search from the road immediately after lastServed and select the first road that matches the highest density.
5. Serve the selected road and mark it as served after its complete signal sequence.
6. Repeat until no unserved occupied road remains; then perform a fresh traffic scan.

6.5 Signal Timing Algorithm

Density	Green	Yellow	All Red
LOW	7 s	3 s	1 s
MEDIUM	10 s	3 s	1 s
HIGH	15 s	3 s	1 s

6.6 Simplified Pseudocode

```
INITIALIZE sensors, LEDs, LCD
SET all roads RED

LOOP:
  read and filter all 12 sensors
  density[N,S,E,W] = number of active sensors on each road

  if starting a new service cycle:
     snapshot densities
     clear served flags

  choose highest-density unserved road
  if tied, rotate priority after lastServed

  GREEN selected road for 7/10/15 s
  YELLOW selected road for 3 s
  ALL RED for 1 s
  mark selected road served

  if another occupied unserved road exists:
     serve next road
  else:
     begin a new scan cycle
```

6.7 Timing and Display Update Parameters

Parameter	Value
Sensor samples per reading	7
Detection majority threshold	4 of 7 samples
LCD update interval	200 ms
LCD page change interval	2500 ms
Serial print interval	500 ms
LOW green	7000 ms
MEDIUM green	10000 ms
HIGH green	15000 ms
Yellow	3000 ms
All-red safety gap	1000 ms

6.8 Summary

The software implementation converts twelve filtered sensor inputs into four density values, schedules roads by priority while avoiding repeated service within the same cycle, and controls the traffic LEDs through a safe state machine. The next chapter documents the functional tests and observed results.

CHAPTER 7

EXPERIMENTAL RESULTS

7.1 Test Setup

The completed prototype was tested with the Arduino Mega, twelve IR sensors, twelve traffic LEDs and the 16 x 2 I2C LCD connected simultaneously. Toy vehicles / objects were placed in different sensing zones to create EMPTY, LOW, MEDIUM and HIGH traffic conditions. Serial Monitor output was used to confirm sensor patterns and calculated density while the physical LEDs and LCD were observed.

Test Condition	Configuration
Controller	Arduino Mega 2560
IR sensors	12 (3 per approach)
Traffic indicators	4 x Red, 4 x Yellow, 4 x Green LEDs
Display	16 x 2 I2C LCD at address 0x27
Serial Monitor	9600 baud
Power	Stable 5 V / USB supply used for prototype testing
Traffic source	Toy vehicles / test objects placed in sensing zones

7.2 Sensor Density Test Results

Sensor Pattern	Expected Density	Observed Behavior
000	EMPTY	No green request from that road - PASS
100	LOW	7 s green when selected - PASS
010	LOW	7 s green when selected - PASS
001	LOW	7 s green when selected - PASS
110	MEDIUM	10 s green when selected - PASS
101	MEDIUM	10 s green when selected - PASS
011	MEDIUM	10 s green when selected - PASS
111	HIGH	15 s green when selected - PASS

The order-independent classification solved the earlier issue where patterns such as 101 were treated as invalid. The final code correctly classifies density using the total number of active sensors.

7.3 Adaptive Green Timing Test

Density	Programmed Green Time	Result
LOW	7 seconds	PASS
MEDIUM	10 seconds	PASS
HIGH	15 seconds	PASS
EMPTY	No green phase requested	PASS

7.4 Individual Approach Test Results

Approach	Sensors	Signal Pins	Result
North	22, 23, 24	34, 35, 36	PASS
South	25, 26, 27	37, 38, 39	PASS
East	28, 29, 30	40, 41, 42	PASS
West	31, 32, 33	43, 44, 45	PASS

7.5 Mixed-Density Priority Tests

Test Case	Traffic Condition	Expected / Observed Service
1	N=HIGH, S=LOW, E=MEDIUM, W=EMPTY	HIGH first, then MEDIUM, then LOW; empty road skipped - PASS
2	N=HIGH, S=HIGH, E=LOW, W=MEDIUM	Two HIGH roads served before MEDIUM and LOW; tie handled by rotating priority - PASS
3	N=LOW, S=LOW, E=LOW, W=LOW	All four receive one turn in fair rotating order - PASS
4	N=HIGH, S=HIGH, E=HIGH, W=HIGH	All four receive 15 s green turns with safe transitions - PASS

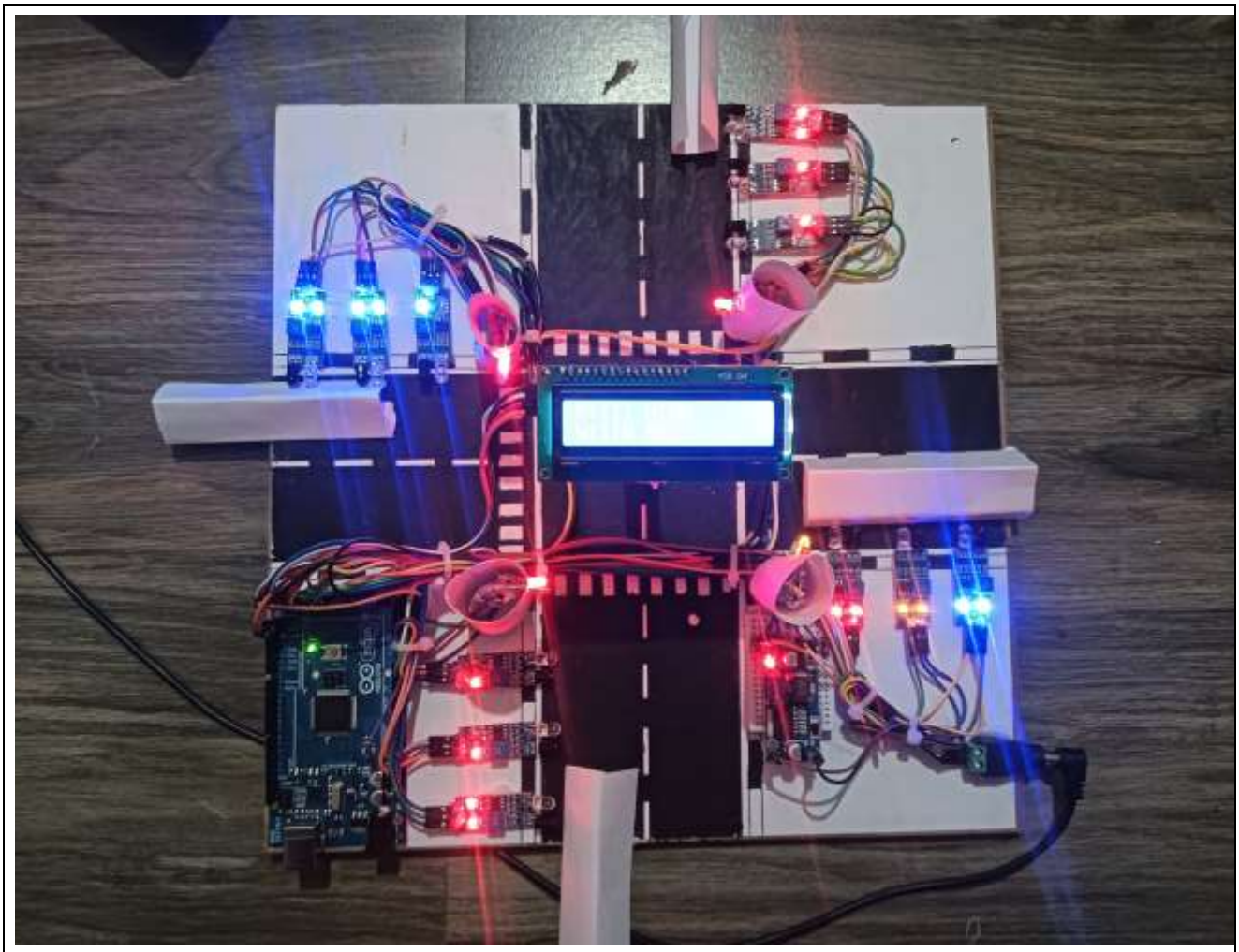
7.6 Signal Safety Test

Check	Expected Result	Observed
One green at a time	Only selected road GREEN	PASS
Other roads during green	Remain RED	PASS
Transition after green	Selected road YELLOW for 3 s	PASS
Intersection clearance	ALL RED for 1 s	PASS
Next road	Selected after all-red interval	PASS

7.7 LCD Display Results

Display Test	Example Output	Result
Overall density	N:H S:L E:M W:0	PASS
Active green timer	N GREEN 15s	PASS
Yellow timer	E YELLOW 3s	PASS
All-red timer	ALL RED 1s	PASS
North/South detail	N:101 MED / S:001 LOW	PASS
East/West detail	E:110 MED / W:111 HIGH	PASS

7.8 Final Hardware Setup Photo



7.9 Programmed Performance Parameters

Metric	Value
Sensor filtering	7 samples, majority threshold 4
LCD refresh	Every 200 ms
LCD page interval	2.5 s
Serial status interval	500 ms
Low green	7 s
Medium green	10 s
High green	15 s
Yellow	3 s

Metric	Value
All red	1 s
Traffic scheduling	Highest density first + rotating tie handling

7.10 Summary

The prototype successfully demonstrated four-way density detection, order-independent sensor classification, adaptive signal timing, safe phase transitions, priority scheduling and LCD monitoring. The staged integration approach - North, North+South, North+South+East, then all four roads - helped isolate wiring and software issues during development.

CHAPTER 8

APPLICATIONS, ADVANTAGES AND DISADVANTAGES

8.1 Applications

8.1.1 Education and Embedded Systems

Application	Description
Embedded Systems Laboratory	Demonstrates digital inputs, outputs, I2C, state machines and timing
Mini / Final-Year Project	Demonstrates adaptive control using low-cost hardware
Algorithm Demonstration	Shows density priority, tie handling and service cycles

8.1.2 Traffic and Access-Control Prototypes

Application	Description
Smart intersection model	Prototype study of demand-based signal timing
Campus / industrial gate model	Demonstrates priority according to queue occupancy
Parking exit control	Can be adapted to choose between multiple vehicle entry/exit lanes
Research test platform	Can be expanded with cameras, wireless monitoring or emergency priority

8.2 Advantages

Advantage	Description
Adaptive Timing	Green time changes with detected traffic density
Priority Control	Higher-density roads are served before lower-density roads
Fair Tie Handling	Rotating selection avoids permanent fixed priority
Empty Road Skipping	Roads with no detected traffic are not given an unnecessary active turn
Low Cost	Uses commonly available Arduino and IR modules
Simple Debugging	Direct Mega pins make wiring and troubleshooting straightforward
Safety Sequence	Yellow and all-red intervals are enforced before the next green
Live Monitoring	LCD and Serial Monitor show traffic and system state
Order-Independent Density	Any one/two/three active sensors map to LOW/MEDIUM/HIGH

8.3 Disadvantages and Mitigation

Disadvantage	Description	Possible Improvement
IR sensitivity	Detection can vary with target surface, distance and ambient conditions	Improve mounting, shielding and calibration
Zone-based estimate	System estimates occupied sensing zones rather than counting exact vehicles	Use camera vision, ultrasonic or radar sensors
Prototype scale	Small road model does not represent full real-world intersection geometry	Use industrial detectors and certified controllers
No pedestrian input	Prototype controls vehicle approaches only	Add pedestrian button and pedestrian signal phase

Disadvantage	Description	Possible Improvement
No emergency priority	Ambulance / fire-vehicle priority is not implemented	Add RFID, V2X or authenticated emergency input
No remote monitoring	Status is local to LCD / Serial Monitor	Add ESP32, Wi-Fi or cloud dashboard
Fixed density timing values	7/10/15 s values are preconfigured	Tune timings from measured traffic flow or adaptive optimization

CHAPTER 9

CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

The Density Based Traffic Light Control System was successfully designed, implemented and tested as a four-way adaptive traffic prototype. The Arduino Mega 2560 reads twelve IR sensing zones, estimates traffic density independently for North, South, East and West, and controls four red-yellow-green signal sets according to the detected demand.

The final implementation classifies density using the number of active sensors rather than a strict physical order. This provides robust prototype behavior: any one active sensor is LOW, any two are MEDIUM and all three are HIGH. The controller assigns 7 s, 10 s or 15 s of green time, followed by a 3 s yellow phase and 1 s all-red safety gap. Higher-density approaches are served first, while rotating tie handling improves fairness when densities are equal.

Objective	Status
Four-way Arduino Mega traffic controller	ACHIEVED
12 IR sensor integration	ACHIEVED
EMPTY / LOW / MEDIUM / HIGH density classification	ACHIEVED
Order-independent density detection	ACHIEVED
7 / 10 / 15 second adaptive green timing	ACHIEVED
3 second yellow transition	ACHIEVED
1 second all-red safety interval	ACHIEVED
Highest-density priority	ACHIEVED
Fair tie handling	ACHIEVED
16 x 2 LCD live display	ACHIEVED
Serial diagnostic output	ACHIEVED

Key Learnings:

1. Digital sensor interfacing and calibration using IR obstacle sensors.
2. Stable input processing using repeated sampling and majority decision.
3. Adaptive control based on real-time input conditions.
4. State-machine based signal sequencing using millis() timing.
5. I2C LCD interfacing and live status display.
6. Priority scheduling and fairness handling for equal traffic conditions.
7. System integration using staged hardware and software testing.

The completed system is functional and suitable for academic demonstration of density-aware traffic control. It should be presented as a prototype model, not as a replacement for certified real-world traffic-signal equipment.

9.2 Future Scope

Enhancement	Description	Priority
Camera-Based Detection	Use computer vision for vehicle counting and lane occupancy	High

Enhancement	Description	Priority
Emergency Vehicle Priority	Provide authorized priority to ambulance / fire vehicles	High
Pedestrian Phase	Add pedestrian buttons, timers and walk signals	High
IoT Dashboard	Send traffic density and signal status to a web/cloud dashboard	Medium
Data Logging	Store traffic history on SD card or cloud database	Medium
Dynamic Timing Optimization	Calculate green time from measured queue length and arrival rate	Medium
Wireless Sensor Nodes	Reduce wiring using distributed sensor nodes	Medium
Solar / Backup Power	Improve independence and power resilience	Low
RTC Timestamping	Add date/time to traffic logs	Low
Real Hardware Signals	Drive higher-power lamps through proper isolated driver circuits	Low

Recommended Next Steps:

1. Mount all sensors and signals permanently on the road model and re-calibrate each IR sensor.
2. Add an ESP32 or Wi-Fi module for a live web dashboard if remote monitoring is required.
3. Add pedestrian and emergency-vehicle inputs for a more complete intelligent intersection model.
4. Collect traffic data and tune the LOW/MEDIUM/HIGH green durations using measured queue-clearing performance.

REFERENCES

1. Arduino, Arduino Mega 2560 Rev3 - product documentation and board reference.
2. Microchip Technology, ATmega2560 / ATmega2560V Microcontroller Datasheet.
3. Arduino Wire Library Documentation - I2C / TWI communication.
4. LiquidCrystal_I2C Library Documentation - I2C character LCD interface.
5. Hitachi, HD44780U LCD Controller / Driver Datasheet (reference for 16 x 2 character displays).
6. NXP Semiconductors, PCF8574 Remote 8-bit I/O Expander for I2C-bus (commonly used on LCD backpacks).
7. IR obstacle sensor module manufacturer / supplier documentation for the specific modules used in the prototype.
8. Arduino IDE Documentation - sketch compilation, board programming and Serial Monitor usage.
9. Project source code and experimental observations from the completed Density Based Traffic Light Control System prototype.